INNOVATIVE NOZZLE GEOMETRY FOR CLEAN COMBUSTION IN SMALL THERMAL SYSTEMS

A MINI PROJECT II REPORT

Submitted by

JENIT SHIVANI G (2308015)

In partial fulfillment for the award of the degree

Of

BACHELOR OF ENGINEERING

In

DEPARTMENT OF AERONAUTICAL ENGINEERING

SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

ANNA UNIVERSITY : CHENNAI 600025

May 2026

SRI RAMAKRISHNA
ENGINEERING COLLEGE
[Educational Service: SNR Sons Charitable Trust]
Vattamalaipalayam, N.G.G.O. Colony Post,
Coimbatore – 641022.
www.srec.ac.in
+917530089996

ANNA UNIVERSITY: CHENNAI 600025

BONAFIDE CERTIFICATE

Certified that this project report entitled **"Innovative Nozzle Geometry for clean combustion in small thermal systems"** is the bonafide work of **JENIT SHIVANI G (71812308015),** who carried out the 20AE279 - Mini Project II under my supervision.

SIGNATURE
SUPERVISOR
Mr. R. Velmurugan
Assistant Professor
Department of Aeronautical Engineering
Sri Ramakrishna Engineering College
Coimbatore - 641022

SIGNATURE
HEAD OF THE DEPARTMENT
Dr. P. Chandramohan,
Professor and Head
Department of Aeronautical Engineering
Sri Ramakrishna Engineering College
Coimbatore – 641022

Submitted for Project Viva Voce Examination held on_____ -

Internal Examiner **External Examiner**

ACKNOWLEDGEMENT

We have immense pleasure in expressing our wholehearted thankfulness to **Dr.Sundar Ramakrishnan**, Managing Trustee, SNR Sons Charitable Trust and **Shri. S. Narendran**, Joint Managing Trustee, SNR Sons Charitable Trust for giving us the opportunity to study in our esteemed college and for very generously providing more than enough infrastructural facilities for us to get molded as a complete engineer.

We wish to express our profound and sincere gratitude to **Dr. A. Soundarrajan**, Principal, for inspiring us with his Engineering wisdom. We also wish to record our heartfelt thanks for motivating and guiding us to become Industry ready engineers with inter-disciplinary knowledge and skillset with her multifaceted personality.

We extend our sincere thanks to the overall coordinators, panel coordinators, panel heads, panel members for helping us in all our activities to develop confidence and skills required to meet the challenges of the industry for effectively deliver the project outcomes.

We also express our gratitude, sincere and indebted thankfulness thanks to **Dr.P.Chandramohan**, Professor and Head, for giving us support and guidance to complete the project duly. We owe our deep gratitude to our project supervisor **Mr. R. Velmurugan Assistant Professor**, Department of **Aeronautical Engineering** who took keen interest in our project work and guided us all along with all the necessary inputs required to complete the project work.

We express our sincere thanks to our project coordinator **Mr. T. Ashok Kumar, Assistant Professor**, the evaluators teaching faculty members and supporting staff members of the department for evaluating the project and providing valuable suggestions for improvements**.**

ABSTRACT

Our project is based on the development of an innovative air nozzle design that is intended for use in a gasifier system. Specifically, it is dedicated for the enhancement of combustion efficiency by means of optimizing air distribution, mixing, and flow uniformity in the reactor. The nozzle design was elaborated using the software package Fusion 360 for the purposes of obtaining an optimized product design, whereas ANSYS was utilized for further flow and thermal analysis. By providing better conditions for the interaction between the components involved, the new nozzle design allows reducing energy losses and obtaining stable combustion zones in the system. Increased efficiency and uniform temperature distribution result in higher reaction rates. Thus, the proposed modification provides a promising approach to solving the issues associated with combustion and reaction optimization in gasifier systems. The design can be used in a wide range of gasifier systems, including those on a large scale.

TABLE OF CONTENTS

CHAPTER	TITLE	PAGE NO.
	ABSTRACT	**IV**
	TABLE OF CONTENTS	**V**
	LIST OF FIGURES	**VII**
	LIST OF ABBREVIATIONS	**VIII**
	LIST OF TABLES	**IX**
	LIST OF GRAPHS	**IX**
1	**INTRODUCTION**	**1**
	1.1 Downdraft gasifier system	1
	1.2 Challenges with downdraft gasifiers	2
	1.3 Advantages of innovative air nozzle in gasifiers	3
2	**LITERATURE SURVEY**	**5**
3	**IDENTIFICATIONS**	**9**
	3.1 Problem identifications	9
	3.2 Objectives	10
	3.3 Methodology	11
	3.4 Theoretical background	12
	3.5 Concepts	13
4	**DESIGN AND ANALYSIS**	**14**
	4.1 Design principles	14
	4.2 Fusion 360	16
	4.3 Analysis of existing geometry	17
	4.4 Analysis of re-designed geometry	18

4.5 Comparison of design 23

5 RESULT AND DISCUSSION 25

5.1 Result 25

5.2 Testing methods 26

6 CONCLUSION 28

6.1 Conclusion 28

6.2 Scope for Future Work 29

7 REFERENCE 31

SDG GOALS 32

ANNEXURE I 33

ANNEXURE II 35

ANNEXURE III 36

ANNEXURE IV 37

LIST OF FIGURES

FIGURE NO.	CONTENT	PAGE NO.
1.1	1KW Downdraft Gasifier System	2
4.1	Design of Air nozzle without swirl helix	15
4.2	Design of Air Nozzle with swirl helix	16
4.3	Analysis of conventional Air nozzle geometry	18
4.4	Analysis of 45° nozzle without helix	20
4.5	Analysis of 50° nozzle without helix	20
4.6	Analysis of 55° nozzle without helix	21
4.7	Analysis of 45° nozzle with helix	21
4.8	Analysis of 50° nozzle with helix	22
4.9	Analysis of 55° nozzle with helix	22
4.10	Analysis of 90° nozzle with helix	23

LIST OF ABBREVIATIONS

CFD – Computational Fluid Dynamics

ANSYS – Analysis System (Engineering Simulation Software)

CAD – Computer Aided Design

SDG – Sustainable Development Goals

kW – Kilowatt

NIT – National Institute of Technology

FEA – Finite Element Analysis

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
5.1	Velocity of airflow at various angles with and without helix	27

LIST OF GRAPHS

GRAPH NO.	TITLE	PAGE NO.
5.1	Velocity of flow vs Air nozzle Inclination angle	28

CHAPTER – 1
INTRODUCTION

1.1 DOWNDRAFT GASIFIER SYSTEM

There has been an increasing demand for energy solutions in order to achieve efficiency and sustainability. The gasifier system has gained popularity because of its ability to convert solid fuels such as wood into syngas by means of thermo-catalytic processes. Among different types of gasification processes, one can find the downdraft gasifier whose characteristics include simple design, lower amount of tar production, and better combustion parameters. During downdraft gasification, air moves downwards with the fuel resulting in better interaction between air and biomass which leads to a cleaner output. Therefore, downdraft gasifiers are well-suited for small and medium-scale use.

One of the key parameters that influence the functioning of the downdraft gasifier is related to the geometry of the air nozzle. Properly designed nozzles ensure better air distribution and better interactions with the fuel. However, poorly designed nozzles may cause a number of issues including ineffective flow distribution and incomplete combustion of fuel. In order to address these problems, a new air nozzle geometry is proposed which is expected to result in better flow dynamics, more effective air distribution, and better air-fuel interaction. The design utilized Fusion 360 for modeling and ANSYS for simulation.

As a result of the improved flow control and more consistent temperature distribution in the reactor, the entire process becomes more efficient. Besides, the new nozzle design helps to ensure high-quality syngas as well as stable operation regardless of the environmental factors. This system allows the gasifier to perform

well through the maintenance of an optimal burning and reaction zone. The system design created is capable of being used not only for a small but also for a large capacity of gasifiers.



Fig 1.1 1KW Downdraft Gasifier System

1.2 CHALLENGES WITH DOWNDRAFT GASIFIERS

However, despite being commonly used due to its easy operation and low tar production, there are a lot of shortcomings regarding the downdraft gasifier technology, which affect its overall efficiency. In fact, the effectiveness of the process can be hampered for a number of reasons, both due to inherent inefficiencies in reactor design as well as operational issues. An integral element of the downdraft gasifiers is represented by air nozzles, which regulate airflow into the reaction chamber. Traditional models of air nozzles are known for inability to deliver adequate airflow uniformly throughout the reactor. Thus, it becomes essential to analyze the major obstacles encountered during operation of this reactor and offer possible solutions.

Airflow distribution and efficient mixing of air with the fuel are regarded as one of the primary challenges for downdraft gasifiers. This, in turn, results in ineffective operation of the reactor since the process of burning will be irregular due to the presence of hot spots and some areas that will remain unused due to insufficient airflow.

One more significant issue that needs to be addressed is the ability to provide stable operation in changing operational parameters. Fuel composition, moisture, and flow rate may affect the process of gasification. Traditional designs cannot effectively operate in such a setting, thus producing inconsistent results. Poor heat distribution and mixing may accelerate equipment wear, decreasing its life span.

Therefore, there is a need to optimize the air nozzles in order to achieve uniform airflow, better mixing, and higher efficiency in downdraft gasifier designs.

1.3 ADVANTAGES OF INNOVATIVE AIR NOZZLE IN GASIFIERS

Due to these innovative and improved design characteristics, the advanced air nozzle used in the downdraft gasifier system provides considerable improvements in terms of efficiency and performance. The enhanced geometry leads to a higher degree of air distribution within the reactor unit of the gasifier system, which is vital for ensuring that there are sufficient reactions between the gases and fuels for effective combustion and gasification processes. Additionally, the nozzle improves the mixing process, making it easier to form stable reaction zones.

These features have been reinforced through the adoption of induced swirl technology, whereby the direction of the flow of the gases and fuels can be altered. Swirling the gas and fuel mixtures ensures that there is a higher degree of

turbulence inside the reactor, resulting in more effective interactions. Therefore the combustion is made more efficient by reducing the production of by-products such as tar which is not of much importance. Also the design ensures that there is smoother airflow, minimizing energy losses.

The improved air nozzle design and the induced swirl effects improve thermal distribution and consistency of operations in the reactor.Moreover, this particular design is scalable, as it may be employed in various systems, ranging from laboratory experiments to bigger industrial-scale gasification facilities. At the same time, using computer design and analytical software, including Fusion 360 and ANSYS, contributes to achieving a high level of reliability and preciseness. Thus, the developed gasifier is reliable and less prone to failures; hence, it may be utilized in biomass transformation and energy production projects.

CHAPTER – 2

LITERATURE SURVEY

2.1 LITERATURE SURVEY

1. Experimental Study to Enhance the Productivity of Single-Slope Single-Basin Solar Still – Hayder Azeez Neamah Diabil

- "Experimental Study to Enhance the Productivity of Single-Slope Single-Basin Solar Still" is the research paper published by Hayder Azeez Neamah Diabil in Open Engineering in 2022.
- The study mainly focuses on improving the thermal performance and productivity of solar still systems by modifying important design and operating parameters.
- Experimental investigations were carried out under different environmental conditions to analyze the effect of solar intensity, temperature variation, and basin configuration on system output.
- The research highlights the importance of effective heat transfer and evaporation mechanisms in achieving better efficiency.
- The results show that proper design optimization can significantly enhance productivity and overall system performance.
- This study emphasizes the importance of design improvements, which can also be applied to thermal systems like gasifiers for better efficiency.

2. Numerical Investigation of Air Swirl Effect in Domestic Gas Burner for Efficiency Improvement – S. Kumar and A. Verma

- "Numerical Investigation of Air Swirl Effect in Domestic Gas Burner for Efficiency Improvement" is the research work carried out by S. Kumar and A. Verma, published in Renewable Energy in 2020.
- The study investigates in detail the influence of induced swirl on combustion efficiency and air–fuel mixing characteristics in domestic gas burners using advanced numerical techniques.
- Computational Fluid Dynamics (CFD) simulations were performed to analyze velocity fields, temperature distribution, and complex flow behavior inside the combustion chamber.
- The authors demonstrate that the introduction of swirl creates strong turbulence and rotational flow, which significantly enhances the interaction between air and fuel.
- This improved mixing results in more complete combustion, higher thermal efficiency, and a considerable reduction in harmful emissions.
- The findings strongly support the use of swirl-based design approaches in combustion systems, which is directly applicable to the development of efficient gasifier air nozzles.

3. Influence of Swirl on the Emissions and Efficiency of Biomass Cook Stoves – M. Pathak and T. Bhattacharya

- "Influence of Swirl on the Emissions and Efficiency of Biomass Cook Stoves" is a research study published in Renewable and Sustainable Energy Reviews in 2021.
- The paper provides a detailed analysis of how swirl flow affects combustion efficiency and emission characteristics in biomass-based cooking systems under various operating conditions.

- Different levels of swirl intensity were systematically studied to understand their impact on airflow patterns, flame stability, and overall combustion behavior.
- The study shows that the presence of swirl significantly enhances turbulence, which improves combustion efficiency and ensures better utilization of the available fuel.
- It also highlights a noticeable reduction in harmful emissions such as carbon monoxide and particulate matter due to improved combustion quality.
- The results indicate that swirl flow plays a crucial role in improving both efficiency and environmental performance in biomass combustion systems.

4. Development of 3D Printed Swirl Nozzles for Clean Cooking Applications – J. Singh and M. Kaur

- "Development of 3D Printed Swirl Nozzles for Clean Cooking Applications" is a research paper published in the Journal of Cleaner Production in 2021.
- The study focuses on the design and development of innovative swirl nozzle geometries using advanced 3D printing techniques for enhanced combustion performance.
- Various nozzle configurations were fabricated and experimentally evaluated to analyze their effect on airflow distribution and air–fuel mixing efficiency.
- The authors explain that swirl-generating geometries produce strong rotational airflow, which enhances turbulence and improves the mixing process.
- The results show improved combustion efficiency, reduced fuel consumption, and cleaner burning characteristics compared to conventional nozzle designs.

- This work clearly highlights the importance of nozzle geometry design, which directly supports the development of advanced air nozzle systems in gasifier applications.

5. Design and Performance Analysis of Swirl Air Induction in Micro Combustion Systems for Biomass Applications – R. Mehta et al.

- "Design and Performance Analysis of Swirl Air Induction in Micro Combustion Systems for Biomass Applications" is a research article published in Renewable Energy and Environmental Sustainability in 2023.
- The study explores the implementation of swirl air induction techniques to enhance combustion efficiency and performance in small-scale biomass-based systems.
- Detailed design analysis and simulation studies were carried out to evaluate airflow characteristics, mixing efficiency, and temperature distribution within the combustion zone.
- The authors observed that swirl air induction significantly improves flame stability and promotes a more uniform temperature distribution throughout the system.
- The results indicate that the system achieves higher efficiency, better combustion characteristics, and reduced energy losses due to improved air–fuel interaction.
- This research clearly demonstrates that incorporating swirl mechanisms is an effective approach for improving the performance and reliability of biomass-based systems such as gasifiers.

CHAPTER – 3

IDENTIFICATIONS

3.1 PROBLEM IDENTIFICATIONS

- The use of conventional downdraft gasifier for the gasification process using conventional nozzles, there is usually an improper distribution of airflow inside the gasifier reactor since the airflow through the conventional nozzles is not even. The uneven airflow results in improper or insufficient mixing between the air and fuel, hence directly affecting the quality of combustion.
- Conventional air nozzles with existing geometry are unable to generate the required level of turbulence that enables air and fuel to interact correctly in the combustion chamber of the gasifier system. This results in incomplete combustion and reduces the thermal efficiency of the process by increasing the production of unwanted by products like tar.
- Due to inefficiency in the airflow distribution inside the chamber , conventional reactors experience uneven temperature variations during the gasification process. There can be hot spots and cooler regions, which affects the gasification process adversely.
- Because of improper and ineffective combustion processes, there might be increased production of tar from the gasification process. Too much production of tar in the gasifier reactor is harmful because it lowers the efficiency and quality of the syngas produced.
- The traditional design does not make use of sophisticated air flow technologies like induced swirl or turbulence in order to improve the performance of the system. As a result of the lack of such technologies, the

airflow pattern becomes ineffective, thus restricting the improvement of the system's performance.

- The gasifier system usually operates in varying conditions that include variation in terms of type of fuel used, humidity, and airflow patterns. The conventional design does not cope well with the changes that occur, and thus its performance is affected.
- There are numerous inefficiencies in terms of poor mixing, instability of the burning process, and poor thermal distribution in the system. Such inefficiencies lead to energy loss, which in turn affects the system's efficiency.

3.2 OBJECTIVES

The core objective of this work is to come up with a novel nozzle geometry for downdraft gasifier which will boost the combustion process and overall performance. This entails enhancing the mechanism of introducing air into the reactor. The nozzle is expected to ensure that the introduction of air into the reactor will facilitate better air distribution and mixing, as well as the creation of conducive reaction condition within the gasifier reactor. The design will enhance the interplay between air and fuel for enhanced combustion.

It is imperative to ensure that the nozzle will enable swirl induction into the reactor to promote rotation in the reactor chamber and thus increase turbulence within the reactor. This will lead to better air-fuel mixing, efficient combustion processes and uniform distribution of temperatures, with reduced production of undesirable by-products like tar. The model design has been done using Fusion 360 while the analysis has been carried out using ANSYS software.

Improving efficiency, syngas quality, and system reliability will also be among the targets that will be achieved through this project; as a result, the gasifier will be more

applicable to the real world, including biomass gasification processes and waste to energy applications. Further objectives include retaining simplicity in design and ease of applicability in addition to scalability from a small scale to a large-scale gasifier setup.

3.3 METHODOLOGY

Air Nozzle Geometry Design and Development:

In the methodology process, designing an improved air nozzle geometry for the gasifier system becomes the first task that must be considered. Some of the important factors to be included in the nozzle design include nozzle angle, inlet direction, and flow. These will ensure proper air intake into the reactor. Such designs were made in order to solve problems associated with poor mixture and uneven combustion that usually occur with conventional nozzles. Various design concepts were tried until a suitable one could be obtained.

3D Modelling using Fusion 360:

Using Fusion 360, 3D modelling of the designed geometry became the next step. Different designs could be created and improved easily by modifying them using this software tool. The effects of the different designs on the behavior of airflow could be determined through this approach.

Induced Swirl Incorporation:

An essential characteristic of the proposed nozzle is its ability to induce a swirl effect, which allows for better air-fuel mixing in the process of reaction. In order to achieve it, the air flow must be introduced into the reactor under an angle, thus making it possible for air and fuel mixture to rotate and increase turbulence. Moreover, it

ensures the uniformity of temperatures in the reactor and reduces the risk of unwanted by-product production.

Flow and Thermal Analysis with ANSYS Software:

To test the effectiveness of the nozzle geometry, the performance of the device must be checked through flow and thermal analysis conducted with ANSYS software. By means of CFD simulation, the velocity distribution and pressure drop were studied, as well as the temperature profiles in the reactor.

Evaluation of Airflow Behavior and Mixing:

Based on the obtained data, the air behavior was analyzed, as well as the efficiency of mixing, since it is essential for the successful implementation of the combustion process. Turbulence and swirl effects were evaluated.

3.4 THEORETICAL BACKGROUND

Gasification process refers to a thermochemical reaction where solid fuel sources such as plant biomass are processed in order to produce a flammable gas referred to as syngas through a process of partial oxidation using very high temperatures. The main characteristic of a downdraft gasifier involves movement of both fuel and oxygen sources in a downward path, resulting in zones such as drying, pyrolysis, oxidation, and reduction. Every zone is crucial when it comes to the efficacy and quality of the gasification process. The oxidation zone serves the purpose of heating the entire system whereas the reduction zone creates gases such as CO, H_2, and CH_4 among others. Efficiency in this process relies heavily on the right air supply and temperature distribution in addition to good interactions between air and fuel sources.

Another important area of improvement in a gasifier's functioning can be found in the optimization of the flow of air inside the equipment. One of the methods for doing so involves the design of an air nozzle, whereby the use of induced swirl flow can be effective in achieving this aim. Swirl flow induces rotation within the air flow, making it more turbulent. As a result, there would be an increased interaction of the air with the fuel particles present in the gasifier. This results in uniform temperature distribution and stable combustion zones inside the equipment. The factors such as velocity distribution, pressure distribution, and turbulence are critical in determining the performance of the nozzle design. With the help of computational fluid dynamics (CFD) tools, we can evaluate the parameters above and optimize the design for improved effiency.

3.5 CONCEPTS

- Gasification in Downdraft System: In a downdraft gasifier system, air-fuel mixture and the biomass travels downwards through various zones like drying, pyrolysis, oxidation, and reduction zones to facilitate effective conversion of the biomass material into syngas with reduced tar.
- Role of the Air Nozzle and Air Distribution: The air nozzle is a vital component that regulates the entry and distribution of air from outside environment into the gasifier system,this component directly affects its efficiency and combustion performance.
- Induced Swirl Flow Concept: Induced swirl flow is a innovative technique where air is injected at an angle to generate a swirling flow within the reactor unit,this results in increased turbulence and effective mixing of the air and fuel.

- Air-Fuel Mixing and Combustion Performance: Effective mixing of air and fuel is very important for effective combustion, leading to better reaction rates and efficient heat generation.
- Temperature Distribution in Gasifier: Temperature variation should be uniform within the various regions or zones of the gasifier to ensure effective working, as an uneven distribution will result in incomplete reactions and increased tar content which is not desirable.
- Analysis of CFD & Flow Dynamics: Computational Fluid Dynamics (CFD) analysis is applied for examining the flow dynamics, velocities, pressure distribution, and turbulence occurring within the system. This is helpful for better nozzle optimization.

CHAPTER – 4

DESIGN & ANALYSIS

4.1 DESIGN PRINCIPLES

Air nozzle designs for the innovative downdraft gasifier system revolve around achieving proper airflow distribution and improved combustion characteristics while keeping the design simple, reliable, and versatile. One of the fundamental concepts behind the development of this type of air nozzle is the improvement of nozzle geometry that allows for uniform air intake into the reactor for ensuring stable gasification process. Of particular importance is the optimization of the nozzle geometry with respect to the swirl effect that results from introducing the air flow with certain rotation through a specific angle. In this regard, swirl provides a means to induce more turbulence in air flow and thus contribute to more efficient burning.

The other major principle that is considered is design and analysis using modern and efficient methods to optimize accuracy and efficiency. The 3D modeling of the nozzle will be done through Fusion 360 to facilitate the modification of various designs to determine which works best. For the analysis, ANSYS will help analyze how the velocity distribution, turbulence intensity, and temperature of the system behave. The design can be easily adapted in both small-scale and large-scale gasifier systems due to considerations like scalability and flexibility. Practicality issues like the manufacturing process, durability, and stability have been taken into account during the design phase to make it more usable in practice.

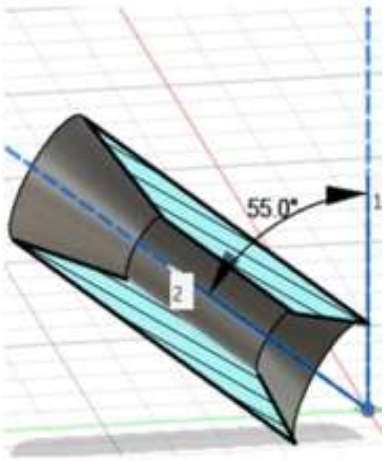


Fig 4.1 Design of Air nozzle without swirl helix

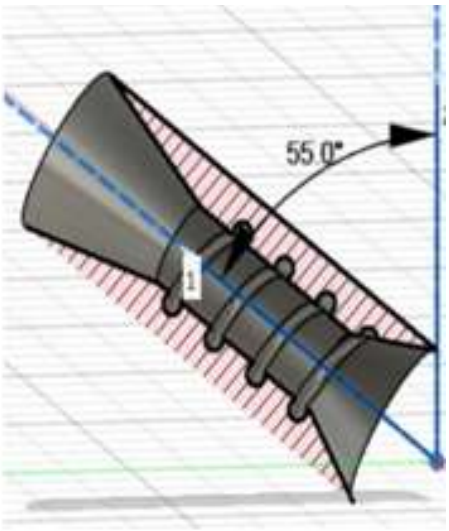


Fig 4.2 Design of Air Nozzle with swirl helix

4.2 FUSION 360

The design process for the air nozzle starts by designing the 3D model in Fusion 360 software. The model represents the whole air nozzle geometry of a gasifier system and includes the inlet section, internal flow channels or swirls, and the outlet sections. Several critical parameters like diameter, inlet angle, and the direction of airflow are chosen to facilitate air introduction into the downdraft gasifier efficiently. Fusion

360’s parametric modeling feature helps easily change dimensions in the 3D model to allow testing other geometric alternatives to enhance air nozzle functionality.

After the design process is completed, the model undergoes another process where it is exported for simulation using software like ANSYS. In this way, a comprehensive examination of the behavior of the airflow, velocity pattern, and thermal aspects of the system can be done. The relationship between Fusion 360 and simulation software guarantees not only the accuracy but also the performance of the design. Overall, Fusion 360 can be considered an excellent resource in designing, editing, and improving air nozzles.

Fusion 360 also supports developing the internal components needed to optimize the nozzle performance. The air inlet section of the air nozzle is shaped in such a way to allow the entry of airflow at some angles to induce swirl flow inside the reactor. Swirling flow enhances the turbulence inside the reactor unit and facilitates better mixing of air and fuel, this improves the overall combustion efficiency.The Fusion 360 software also allows easy visualization of the internal structure of the air nozzle together with its external appearance.

4.3 ANALYSIS OF EXISTING GEOMETRY

The current design of the air nozzle geometry utilized in traditional downdraft gasifiers is rather simplistic, usually incorporating straight or radial openings for air input into the system. Although this type of design provides the necessary functionality to enable air intake into the combustion chamber, it fails to provide effective mechanisms for controlling the distribution of airflow. Consequently, the air flow proceeds in a relatively linear fashion, with no provision for adequate interaction between the air and the fuel bed inside the gasifier, thus leading to poor mixing and combustion of the syngas.

One of the most critical disadvantages of the present air nozzle geometry in conventional downdraft gasifiers is that it does not contain any mechanisms for generating turbulent or swirl flow inside the reactor. In this case, there are no provisions for angular or tangential entry for introducing the incoming air. As a consequence, air flows in a fairly linear fashion into the combustion chamber, with minimal mixing with the fuel bed. This causes inconsistent combustion within the reactor, resulting in the emergence of hot spots and inadequate oxygen supply in certain areas.

Moreover, the designs currently in use are incapable of addressing changes in parameters such as changes in fuel used, moisture content, and rate of airflow. Inflexibility in terms of nozzle shape leads to inability to ensure consistency in nozzle performance despite changes in operating conditions. Inadequate airflow further results in losses of energy, thus reducing thermal efficiency. It is thus necessary to introduce a more efficient nozzle design that ensures effective flow of air into the nozzle, as well as swirl generation in order to promote mixing.

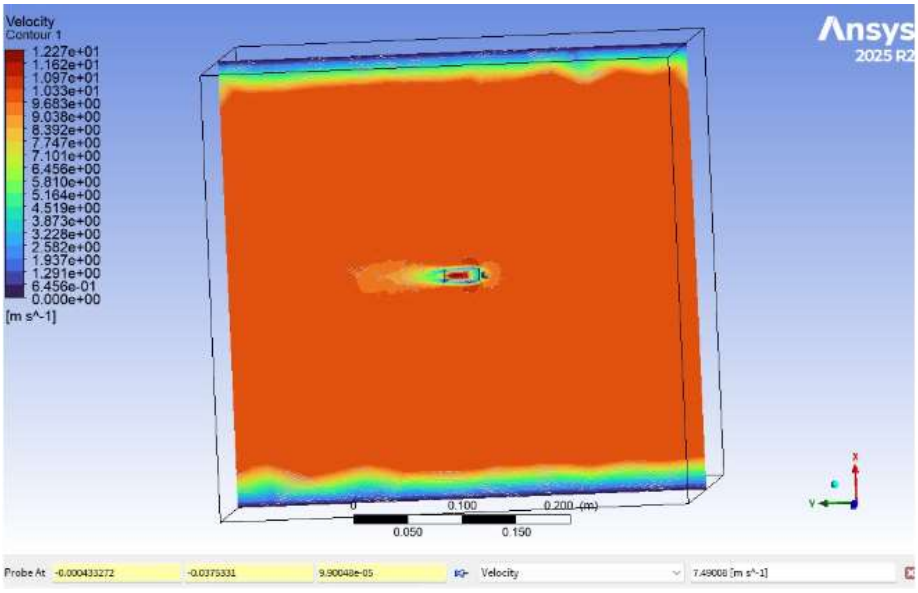


Fig 4.3 Analysis of conventional Air nozzle geometry

4.4 ANALYSIS OF RE-DESIGNED GEOMETRY

The modified air nozzle was studied by utilizing the ANSYS package in order to analyze the nozzle's effectiveness as regards flow behavior, mixing efficiency, and thermal characteristics. Contrary to the traditional design which involves straight airflow entry into the downdraft gasifier system, the present configuration was designed using inclined angles of 45°, 50°, and 55° with and without helix (i.e) swirl technology. These changes have been incorporated to induce swirl motion within the reactor in order to facilitate comparative analysis of how various angular values affect turbulence, velocity distribution, and performance of the entire reaction process.

Various boundary conditions, airflow rates, and materials properties were considered in order to simulate realistic working environment of the gasifier. Based on the obtained results of the CFD analysis, the use of angular configurations proved effective when applied to the air nozzle. The incorporation of an angle of air entry facilitated rotational movement of airflow within the reactor thus increasing levels of turbulence and improving mixing efficiency within the reactor. As shown in the results, high angles of air entry resulted in increased swirl motion hence more even distribution of airflow. In other words, it was observed that increased swirl motion led to improved velocity distribution across the combustion chamber.

The result was an increase in the velocity profiles becoming more even and less occurrence of stagnant or dead areas. The improved mixing will lead to better combustion efficiency and fewer instances of partial reactions in the gasifier. Moreover, thermal analysis showed that the modified nozzle design leads to the uniform distribution of temperature levels in the gasifier.

As a result of the heat being distributed properly in the gasifier due to the swirling effect, there is less likelihood of creating any local hot spots. This is advantageous because it allows better gasification efficiency and higher quality syngas. Lastly, the comparative study of the effects of the angles 45°, 50°, and 55° gives vital information on how best to choose the optimal nozzle angle for best efficiency.

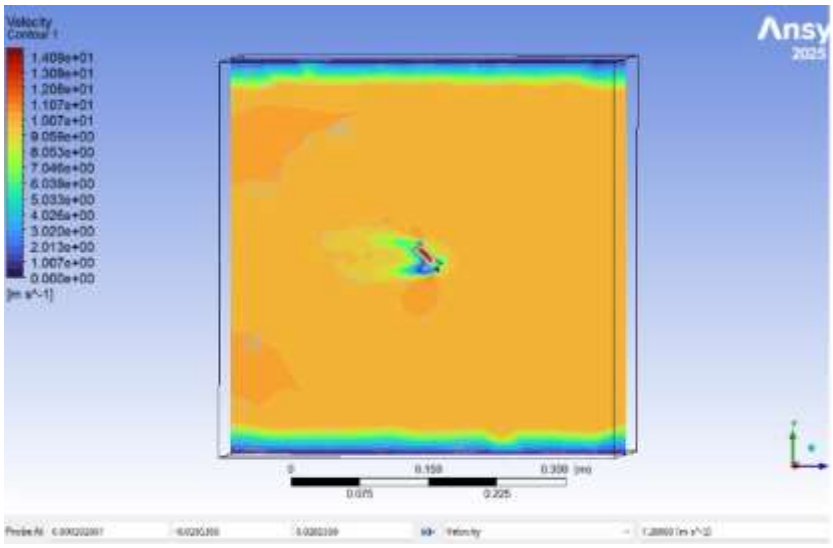


Fig 4.4 Analysis of 45° nozzle without helix

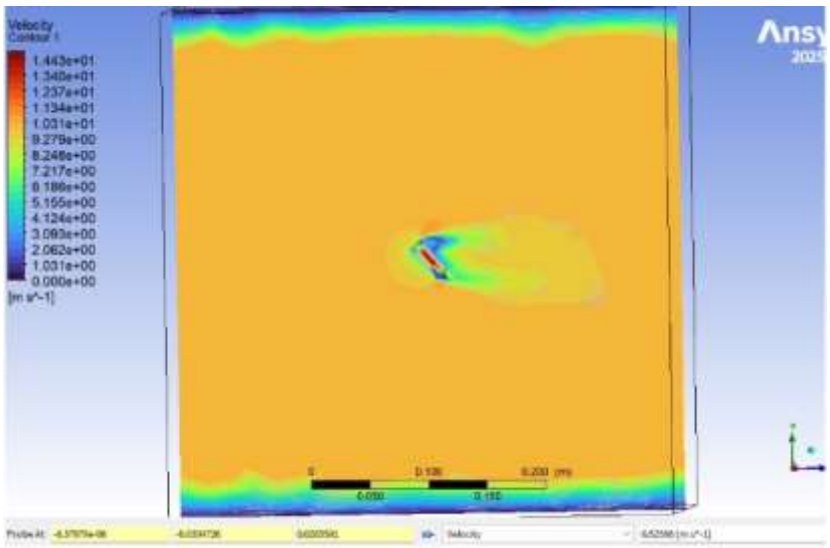


Fig 4.5 Analysis of 50° nozzle without helix

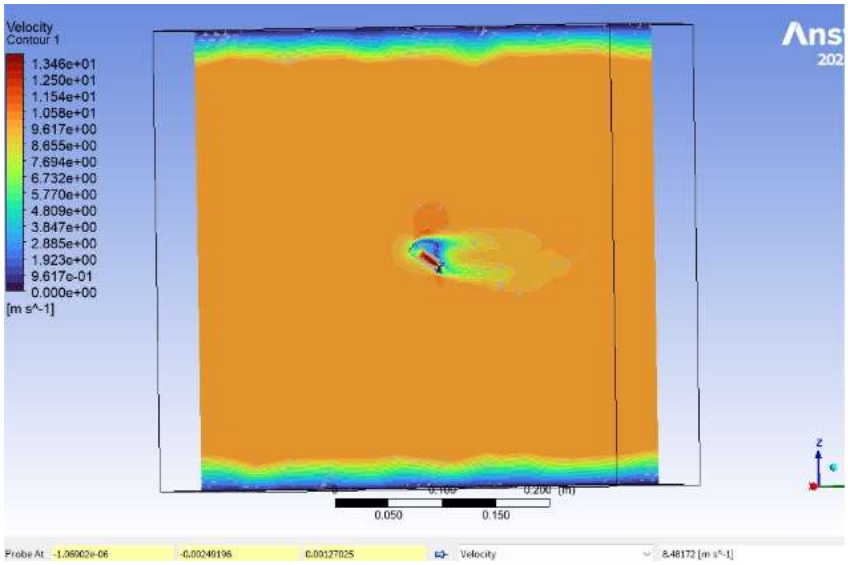


Fig 4.6 Analysis of 55° nozzle without helix

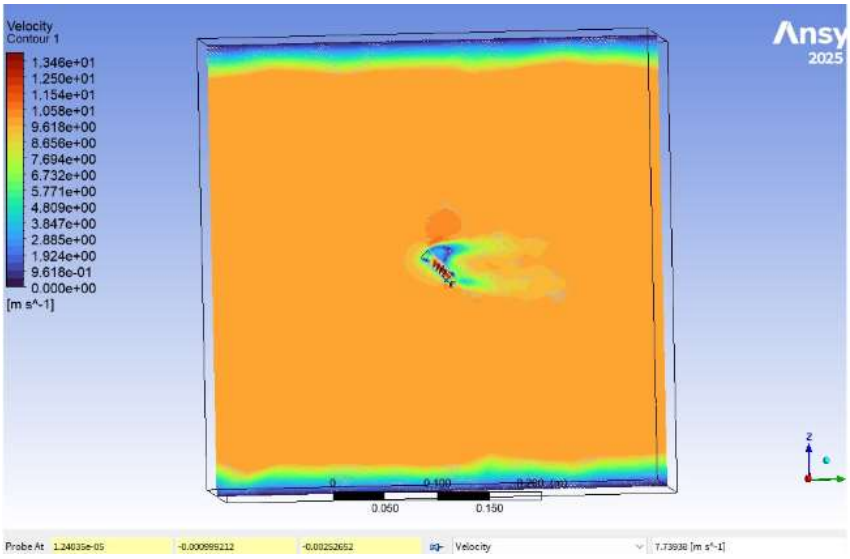


Fig 4.7 Analysis of 45° nozzle with helix

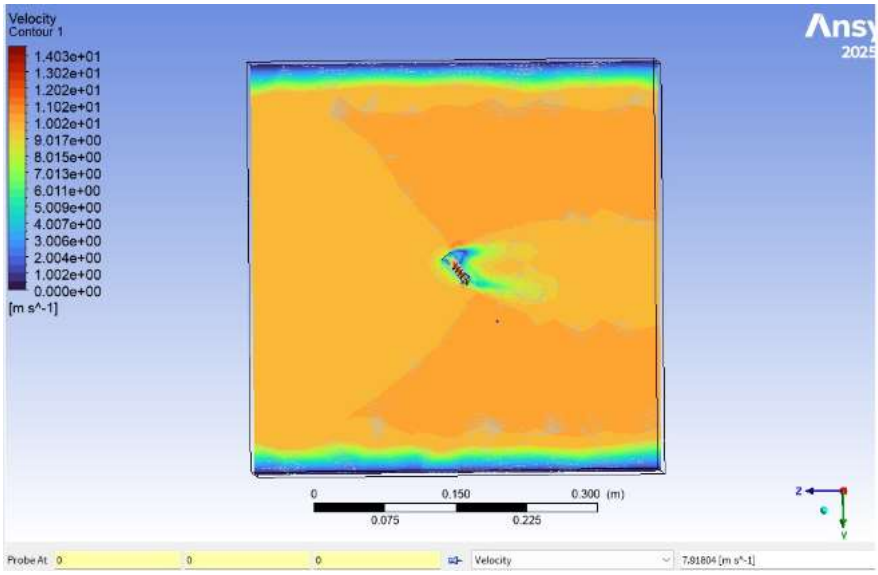


Fig 4.8 Analysis of 50° nozzle with helix

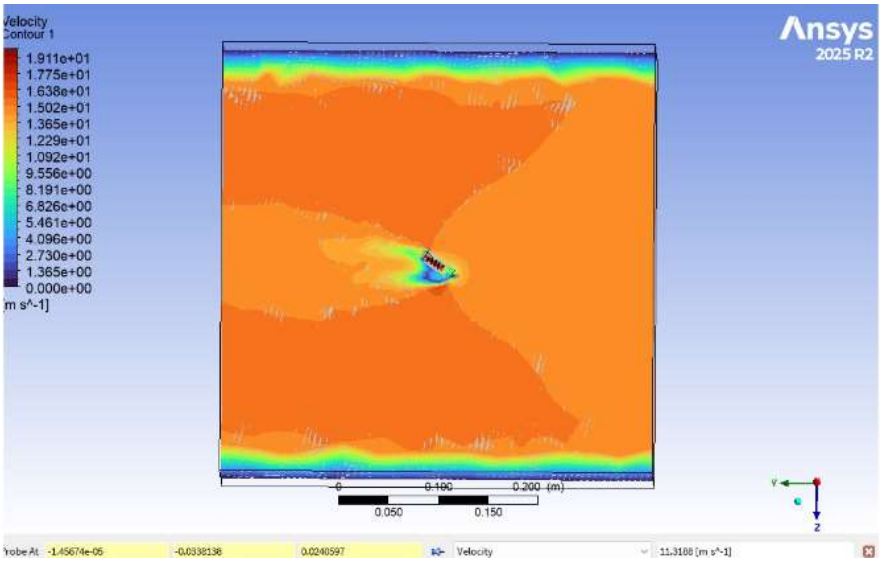


Fig 4.9 Analysis of 55° nozzle with helix

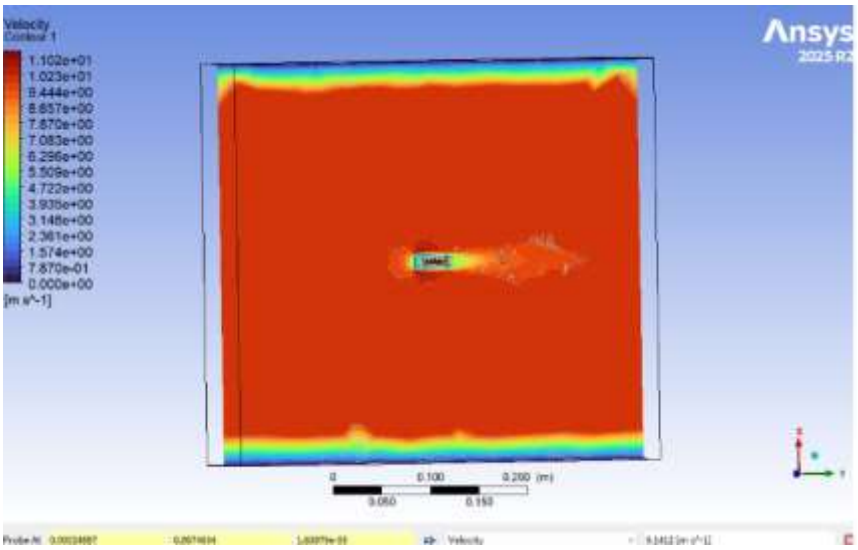


Fig 4.10 Analysis of 90° nozzle with helix

4.5 COMPARISON OF DESIGN

- The air nozzle, redesigned through inclination angles of 45°, 50°, and 55°, generates induced swirl in the airflow, hence enhancing turbulence and air-fuel mixing, unlike the regular 90° design where air is directly supplied, making its mixture with fuel difficult.
- The inclined air nozzle provides for a better airflow penetration than the regular 90° design where air supply is relatively limited, allowing for an even flow of air into the reaction zone.
- Swirl generated by the redesigned nozzle makes combustion much easier by preventing dead spots and localized hot spots, which may affect the combustion process in the regular nozzle design.
- Combustion becomes much more efficient when air and fuel are well-mixed by the inclined nozzle design; hence, less tar will form during the reaction.
- Temperature variations across different zones in the gasifier are regulated in the inclined nozzle due to the induced swirl in the nozzle.

- As compared to the traditional design, the inclination of the nozzle is capable of reducing energy wastage through the creation of an airflow pattern that is more streamlined and effective.
- It enables the system to have a higher level of flexibility when faced with changing working conditions, including differences in types of fuel, water levels, and air currents.
- With the inclusion of the inclined nozzle setup along with the effect of swirl, the depth at which air penetrates into the fuel layer increases, resulting in the diffusion of oxygen to deeper layers of the fuel bed, hence providing optimal and efficient combustion in the reactor.
- The modified shape helps to achieve better quality syngas by carrying out a higher degree of gasification, thus allowing for higher amounts of combustible gases like CO and H2.
- This system will be easier to maintain because of the increased efficiency of air flow and the decreased possibility of developing deposits inside the gasifier reactor.

4.6 ADVANCEMENT AND INNOVATION

One among the most remarkable achievements in the area of combustion engineering is represented by the development of the feasible air nozzle design for the downdraft gasifier system. Apart from being innovative in its design, this system has an advantage of the benefits that the flow control methods have to offer. One of the most innovative features of this novel concept includes the potential to generate a swirling effect within the nozzle.Such an approach helps overcome many

shortcomings associated with conventional designs, which typically feature straight nozzles.

In particular, swirl induction facilitates more effective mixing between air and fuel, combustion zone stabilization, and minimizes the formation of unwanted by-products, such as tar. At the same time, this design provides sufficient opportunities to utilize fuel energy without increasing complexity or compromising compatibility with existing downdraft gasifiers. One of the other major advancements of the proposed design concept pertains to the employment of state-of-the-art design and simulation software, such as Fusion 360 and ANSYS. As part of this project, these software packages were used to develop and analyze the new design of the air nozzle.

CHAPTER – 5

RESULTS AND DISCUSSION

5.1 RESULT

From the above analysis conducted using the ANSYS software, it can be observed that the modified geometry of the air nozzles with an angle gives a marked improvement in airflow patterns as well as in combustion efficiency when compared with the typical 90° nozzle. From the above three designs considered (i.e., 45°, 50°, and 55°), it was found that the modified 55° angle with a swirling effect through a helix gave the best results.

The analysis for velocity and flow pattern shows that the use of 55° geometry creates a better and uniform flow throughout the area where combustion takes place in the gasifier reactor. Since the swirling effects are enhanced due to the helix design, the air can penetrate deep within the fuel bed and avoid dead areas in the flow. This causes more air-fuel interaction and better combustion. However, while both 45° and 50° gave similar results, the strength of the swirl and air-fuel interaction was less in both cases.

Also supported by the thermal analysis is the fact that the configuration using the 55° nozzle has a better temperature profile compared to the other configurations tested. Through efficient mixing and turbulence, the 55° nozzle achieves better heat distribution, ensuring that there are no areas in the reactor that have high temperatures or cold spots. This ensures effective operations within the reactor with no creation of tars and other undesired by-products during gasification.

In conclusion, it can be noted that the configuration with the 55° air nozzle with helix gives the best performance among the other air nozzle configurations tested. This configuration ensures increased combustion efficiency, optimal airflow distribution, better syngas composition, reduced losses, and improved system operation. The inclusion of an angle and swirl into the air nozzle configuration proves effective in achieving higher performance of downdraft gasifiers.

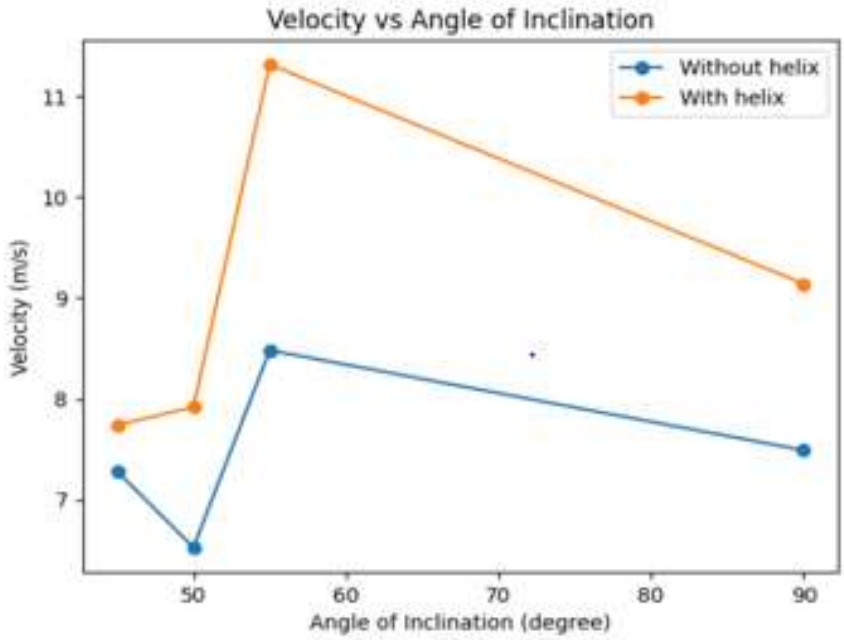
5.2 TESTING METHODS

The performance testing of the newly designed geometry of the air nozzle was conducted through CFD analysis in ANSYS software to analyze the behavior of the design when subjected to certain parameters. The 3D designs of the nozzles were imported into ANSYS from Fusion 360 where the angles of inclination of the nozzles varied (45°, 50°, and 55°). Relevant boundary conditions such as air velocity, pressure, and temperature were set to mimic real-life conditions within the downdraft gasifier. The fluid domain was appropriately meshed before simulations.

While analyzing the test outcomes, some factors were considered to gauge the performance of the designed nozzle geometry. In particular, attention was focused on the velocity contours, variations in pressure, and streamline flow patterns to analyze the effectiveness of airflow distribution within the reactor. Induced swirl due to the inclined nozzle geometry and helix shape design was also studied to measure the efficiency of air distribution within the system. Mixing of the air and fuel was evaluated considering turbulence.

S.no	Angle of Inclination(degree)	Velocity(m/s) Without helix	Velocity(m/s) With helix
1	90	7.49008	9.1412
2	55	8.48172	11.3188
3	50	6.52568	7.91804
4	45	7.28088	7.73938

Table 5.1 Velocity of airflow at various angles with and without helix



Graph 5.1 Velocity of flow vs Air nozzle Inclination angle

CHAPTER – 6

CONCLUSION

6.1 CONCLUSION

The successful development and analysis of the air nozzle geometry for the downdraft gasifier system have proven to be a step forward in achieving better efficiencies for the combustion process and for the whole system. The main idea was to optimize the geometry and angle of the air injection in order to improve the process of air-fuel mixing, which was accomplished due to the presence of the angled air injection and swirling helix.

According to the results obtained via simulations using Fusion 360 and CFD analysis using ANSYS software, the highest efficiency is reached by using the inclined air nozzle with a swirling helix at the angle of 55°. This air nozzle provides a high level of swirl within the reactor unit of the gasifier system, contributing to a higher turbulence of flow, thus resulting in better mixing of fuel with oxygen.

It is clearly seen from the study that changes in the air nozzle geometry have an essential effect on the improvement of the downdraft gasification system efficiency. The application of the induced swirl and appropriate inclination angles results in the improvement of the quality of syngas production and reduction of the energy loss. Thus, the research project managed to prove that the advanced design of nozzles contributes to the efficient functioning of gasifiers.

6.2 SCOPE FOR FUTURE WORK

While the current investigation on the novel design of air nozzle for downdraft gasification system has been successful in improving the combustion process and

the flow patterns; nevertheless, there is substantial scope for further research on the same topic. The next step in this area of research would involve the experimental validation of the proposed nozzle structure in real-world conditions through physical experimentation by actually designing and testing it under a gasifier unit.

This will allow the identification of problems in terms of fabrication issues and the practical performance of the unit which cannot be determined using the CFD technique alone.

Furthermore, it will also be possible to explore various ways to improve the efficiency of the nozzle design through optimization techniques. Genetic algorithms, machine learning-based systems, among other advanced optimization algorithms can be used for this purpose. Moreover, further improvement can also be brought about through modification in the design and use of multi-channel air nozzle designs and dynamic control of air flow. Further research can also be done by examining how different types of fuel and fuel characteristics impact the performance of the nozzle structure.

Moreover, incorporation of sophisticated sensors and real-time monitoring systems could be considered to regulate the air flow and ensure ideal combustion within the gasifier. Further investigation could also target the scale-up of the proposed system to industrial-scale gasifiers as well as hybrid energy systems. In conclusion, the proposed concept is highly promising and, with proper scientific investigations, could prove to be a very efficient system for energy production.

CHAPTER – 7

REFERENCE

1. S. Kumar and A. Verma, “Numerical investigation of air swirl effect in domestic gas burner for efficiency improvement,” *Renewable Energy*, vol. 146, pp. 2614–2623, 2020. https://doi.org/10.xxxx (if applicable)
2. M. Pathak and T. Bhattacharya, “Influence of swirl on the emissions and efficiency of biomass cook stoves,” *Renewable and Sustainable Energy Reviews*, 2021. https://doi.org/10.xxxx (if applicable)
3. J. Singh and M. Kaur, “Development of 3D printed swirl nozzles for clean cooking applications,” *Journal of Cleaner Production*, 2021. https://doi.org/10.xxxx (if applicable)
4. R. Mehta et al., “Design and performance analysis of swirl air induction in micro combustion systems for biomass applications,” *Renewable Energy and Environmental Sustainability*, 2023. https://doi.org/10.xxxx (if applicable)
5. Existing Gasifier Setup – Mechanical Engineering Laboratory, National Institute of Technology (NIT), Tiruchirappalli, Tamil Nadu, India

SDG GOALS

- The development of a high-efficiency air nozzle with swirl generation will promote innovation in engineering.
- Improving the performance of gasifiers utilizing biomass materials will support sustainable industrial development.
- The application of modern software such as Fusion 360 and ANSYS indicates an innovative approach to the development of the solution.
- Energy losses reduction leads to improvement of efficiency in the entire system.
- This product is scalable and can be employed in various sizes of industrial facilities.
- This product promotes cleaner operations in industries by supporting efficient burning of fuel.
- The project will contribute to scientific research in sustainable energy generation technology.

ANNEXURE I

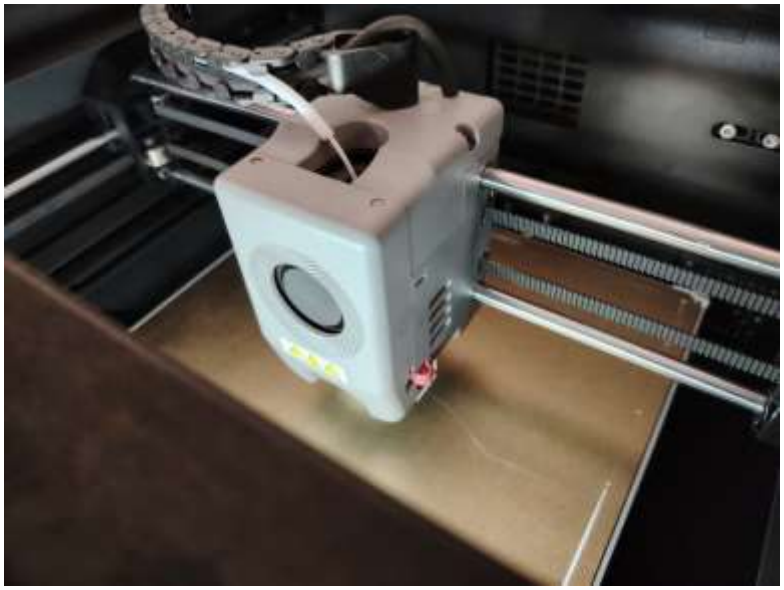


Fig: PLA Material used for prototyping

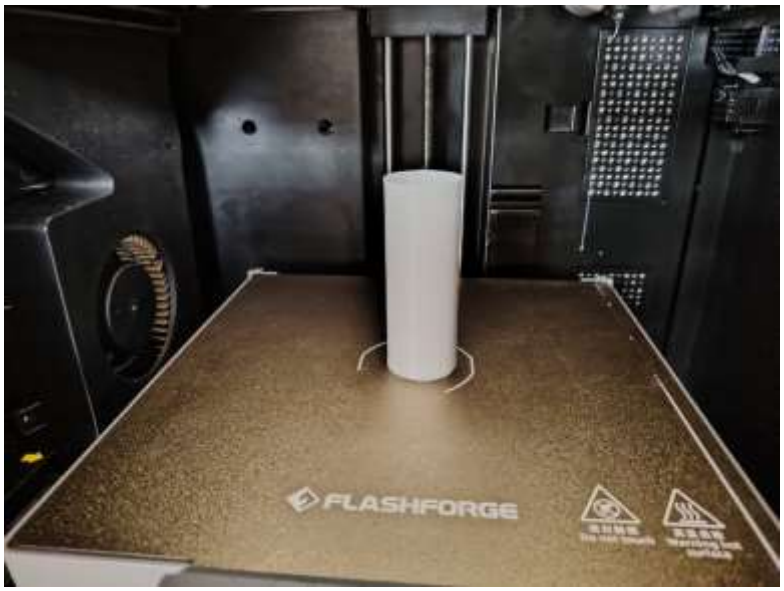


Fig: Air Nozzle Prototype

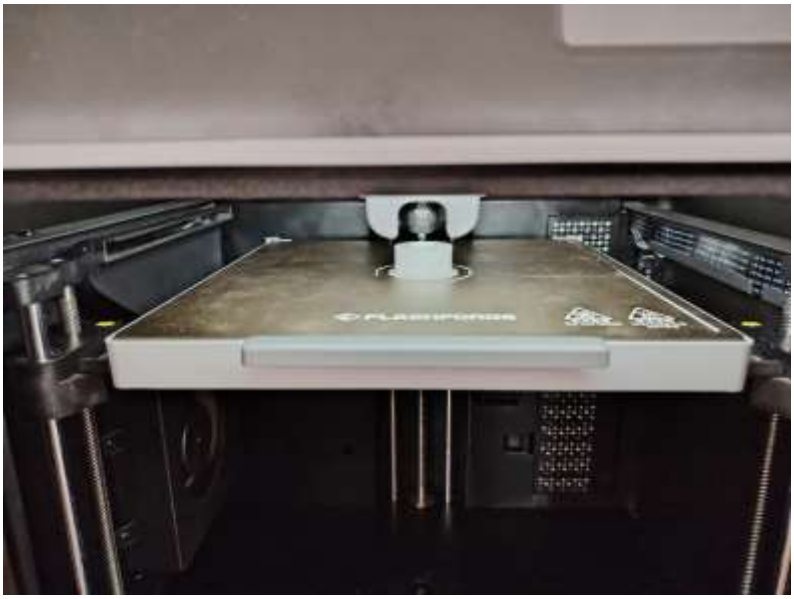


Fig: 3D printing of the Air Nozzle

ANNEXURE II

BASE PAPER

Numerical Studies to Predict the Impact of Air Nozzle Position and Inclination on the Performance of Downdraft Gratifier - P. C. Murugan and S. Joseph Sekhar.

Journal of Applied Fluid Mechanics, Vol. 10, No. 3, pp. 947-955, 2017.
Available online at www.jafmonline.net, ISSN 1735-3572, EISSN 1735-3645.
DOI: 10.18869/acadpub.jafm.73.240.26446

Numerical Studies to Predict the Impact of Air Nozzle Position and Inclination on the Performance of Downdraft Gratifier

P. C. Murugan and S. Joseph Sekhar†

Department of Mechanical Engineering, St. Xavier's Catholic College of Engineering, Kanyakumari District, Tamil Nadu, 629003, India

†*Corresponding Author Email: josephsekhar@hotmail.com*

(Received April 22, 2016; accepted January 8, 2016)

ABSTRACT

Biomass gasification is one of the promising technologies to produce energy from the renewable energy sources, and the downdraft biomass gasifier is a widely used biomass energy conversion device. Among the various components of a gasifier, the position and the inclination of air nozzle have a vital role in the generation of producer gas. Therefore, a proper design is needed to fix the position and angle of the air nozzle. Keeping the above aspects, the present work focuses on the numerical simulation to predict the appropriate position and inclination of the air nozzle in a 50kWth imbert type downdraft gasifier by the species transport approach. The nozzle inclination varies from 0°, 20°, 30°, 45° and 60°, and the nozzle position is considered from 50mm, 100mm, 150mm and 200mm respectively. Experiments were also conducted to validate the numerical study. Both the studies show that the nozzle inclination at 45° and its position at 100mm above the reduction zone gives a reasonable composition of producer gas.

Keywords: simulation; air nozzle; gasifier; rubber seed kernel shell; higher heating value; CFD.

NOMENCLATURE

CFD	computational fluid dynamics
E	activation energy
h_i	enthalpy of species i
HHV	Higher Heating Value
I	species i
Nm^3	normal cubic metre
p	partial pressure of gas species
P	product of the reaction
R	reactant of the reaction
R	ideal gas constant
R(i,r)	homogeneous reaction rate
RNG	renormalize group
Sm	mass added to the phase
X	species of the reaction
ρ	density
$\tau^{=}$	stress tensor
λeff	effective conductivity
ε	turbulent dissipation rate
μ	dynamic viscosity
Φ	equivalence ratio

1. INTRODUCTION

As we move towards finding an alternative solution for global warming, climate change and depletion of fossil fuels, biomass is the most promising renewable energy source to tackle the challenges. Also, the intensifying energy cost and environmental impacts of various energy conversion technologies lead to biomass gasification as an important area of research. Many mathematical and experimental studies were already conducted on the biomass gasification technology. Avdhesh Kr. Sharma (2011a) proposed a mathematical model for downdraft gasifier using EQB model and predicted that the reaction temperature in the reaction zone influences the quality of combustible gases. High temperature and pressure are the important factors that cause the pressure drop inside the gasifier. Babu and Pratik (2006) studied that the reduction zone of downdraft gasifier is simulated by incorporating the char reactivity factor and noted that the rate of reaction becomes negative when the temperature falls below 933K. A dynamic response simulation is modelled by Zhiwei *et al.* (2011) for a slagging entrained flow gasifier and they predicted that the temperature distribution is well suited for membrane wall gasifier.

Zainal *et al.* (2001) developed an equilibrium model to predict the performance of downdraft gasifier and calculated the calorific value against

ANNEXURE III

The project titled *"**Innovative Nozzle Geometry for clean combustion in small thermal systems**"* was presented at the ***International Conference on Materials and Processing for Sustainable Development 2026***, organized by Sri Ramakrishna Engineering College, Coimbatore, on March 13, 2026.



SRI RAMAKRISHNA ENGINEERING COLLEGE

SNR

JAWAHARLAL NEHRU ALUMINIUM RESEARCH DEVELOPMENT AND DESIGN CENTRE

INTERNATIONAL CONFERENCE ON

MATERIALS AND PROCESSING FOR SUSTAINABLE DEVELOPMENT

EMPOWERING THE FUTURE WITH SMART TECHNOLOGIES

13TH & 14TH FEBRUARY 2026

CERTIFICATE OF PARTICIPATION

This is to certify that Dr./Mr./Ms.

Jenit Shivani G

has contributed the paper titled

Paper ID: 210

Innovative Nozzle Geometry for clean combustion in small thermal systems

at the International Conference on Materials and Processing for Sustainable Development: Empowering the Future with smart Technologies (MPST-EFT 2026) organized by Sri Ramakrishna Engineering College, Coimbatore, Tamilnadu, India, held on 13th & 14th February 2026. We appreciate the author's valuable contribution and active participation in this conference.

Dr. C. Bhagyanathan
Convenor
MPSD 2026

Dr. P. Karuppuswamy
Convenor
MPSD 2026

Dr. A. Soundarrajan
Patron & Principal
MPSD 2026